

Midterm Activity 1 - Variables

Instructions

- 1) Carefully analyze the given Java program.
- 2) Trace the references & memory changes step-by-step.
- 3) Complete all required tables & questions.
- 4) Show your logical reasoning where necessary.
- 5) Write your answers in a yellow paper (handwritten).
- 6) Take a clear photo or scan of your work.
- 7) Upload your file in canvas on or before the deadline.
- 8) Submission Format: BSIT2-3_ AREVALO_JAY_CLIFERSON-Code Analysis Activity #1.pdf.

Please review the code below.

```
class Account {
```

```
    String owner;
```

```
    double balance;
```

```
    Account(String owner, double balance) {
        this.owner = owner;
        this.balance = balance;
    }
}
```

```
public class TestAccount {
```

```
    public static void main (String[] args) {
```

```
        Account acc1 = new Account
```

```
        Account acc2 = new Account
```

```
        Account acc3 = acc1;
```

```
        acc2.balance = 9000;
```

```
        acc3.owner = "Miguel";
```

```
        System.out.println(acc1.owner + " " + acc1.balance);
```

```
        System.out.println(acc2.owner + " " + acc2.balance);
```

```
        System.out.println(acc3.owner + " " + acc3.balance);
    }
}
```


Please answer the following questions:

Part A - Variable Identification table

Variable Name	Datatype	Variable Type	Stored in
acc1	Account	Reference / Local	Stack (points to Heap object)
acc2	Account	Reference / Local	Stack (points to Heap object)
acc3	Account	Reference / Local	Stack (points to the same Heap object as acc1)
owner	String	Instance field	Heap (Inside Account object)
balance	double	Instance field	Heap (Inside Account object)

Part B - Explanation - Please see example below.

Statement	What happens?	Memory Impact
1) Account acc1 = new Account("Carks", 5000);	1) This creates a new Account object with owner "Carks" & balance 5000, acc1 which is a local variable that stores the reference	1) This stored onto the Heap, new object allocated. Stack; reference stored.
1) Account acc2 = new Account("Diana", 7000);	1) This is the same as acc1 since it creates another Account object with owner "Diana" with a balance of 7000. acc2 stores the reference	1) Again this another Heap or another object which is allocated. Again it's Stack; reference is stored.
1) Account acc3 = acc1;	1) This just basically points to acc1 & which just basically points to the same object as acc1 & in turn just replicates 1	1) The memory is unchanged since there is no new object since both local variable points to the same Heap object.
1) acc2.balance = 8000;	1) This just updates the balance of acc2	1) Heap it just modifies the object's balance.
1) acc3.owner = "Miguel";	1) Now this changes the owner field of the object reference in acc3 and thus Carks becomes Miguel.	

No. _____

Date _____

Note:

Part C - Output

As you can see it charged the owner of acct 1 entirely.

Miguel 5000.0

Diana 8000.0

Miguel 5000.0